From documents to reviewable relationships

An AI-assisted tool for private-client solicitors and advisers: reconstruct family and trust relationships, inspect their exact documentary Evidence, and decide what to accept.

Alice Chen —— Parent of ——→ **Bob Chen**

Attendance Note - Meeting with Alice Chen

"Alice Chen confirmed that Alice Chen and David Chen are the parents of Bob Chen."

A relationship is useful when the reader can inspect the words behind it.

Product and research case study

Synthetic prototype. No real client data.

Case 01: exact source-backed relationship; editorial diagram.

Reconstruct the facts. Keep the words within reach.

A private-client solicitor or adviser needs a factual starting point: who belongs to the family, who holds a Trust role, and which document supports each connection. The work begins in documentary language, before substantive advice can follow.

The question inside Case 01

> “The purpose of the meeting was to establish an accurate understanding of the current family and beneficiary position before any further review or substantive advice is undertaken.”

Exact passage · Attendance Note – Meeting with Alice Chen

Across documents, the same person may appear under another name; a later note may confirm a fact or describe a changed role. A useful relationship map must preserve those distinctions and the route back to the text. A plausible connection alone is insufficient.

Private Client Graph explores this workflow with synthetic material. Its proposition is an inspectable machine proposal that the professional can review before accepting it as current Matter state. This is an intended use, not evidence of practitioner adoption or time saved.

Reading route Product 3–7 / Method 8–13 / Findings 14–18 / Next 19–20

One person.
Two different claims.

Bob Chen's family connection and his beneficiary status answer different questions. Case 01 states both explicitly. The graph must retain each relationship with its own supporting Evidence; one claim does not establish the other.

Family relationship

Alice Chen ⟶ **Bob Chen** Parent of

> “Alice Chen confirmed that Alice Chen and David Chen are the parents of Bob Chen.”

Trust role

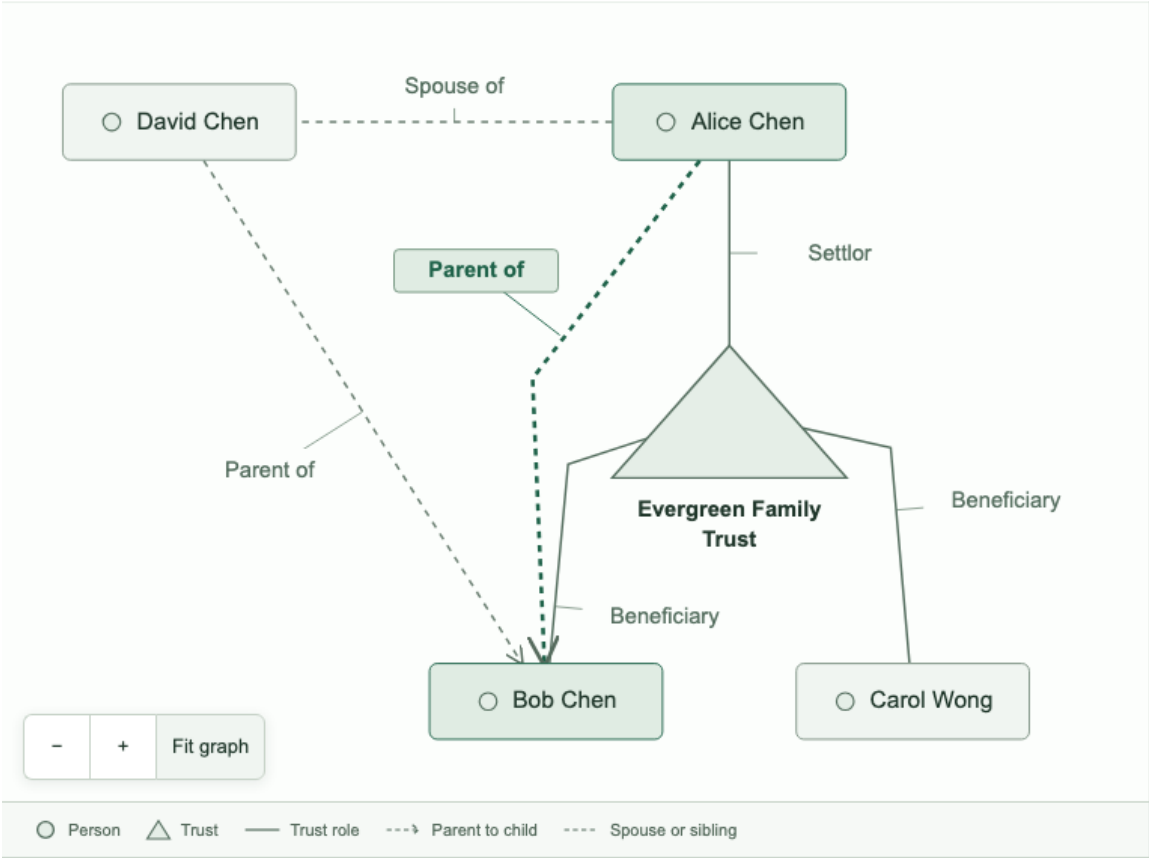
Bob Chen —— **Evergreen Family Trust** Beneficiary of

> “Alice Chen confirmed that Bob Chen is a beneficiary of the Evergreen Family Trust.”

The same note identifies Carol Wong as another beneficiary while recording no family relationship between Carol and the Chen family. A shared Trust connection is therefore not permission to add a family link. Documentary support governs what can be proposed.

A graph you can question

A summary can describe a family without making each claim easy to check. PCG lets the practitioner select a relationship, read its Evidence, and locate the exact passage in its Source.



Actual practitioner workspace: selected relationship detail

Select the claim

Alice Chen is the parent of Bob Chen. Family relationships and Trust roles remain distinct claims.

The selected connector leads to the quotation below.

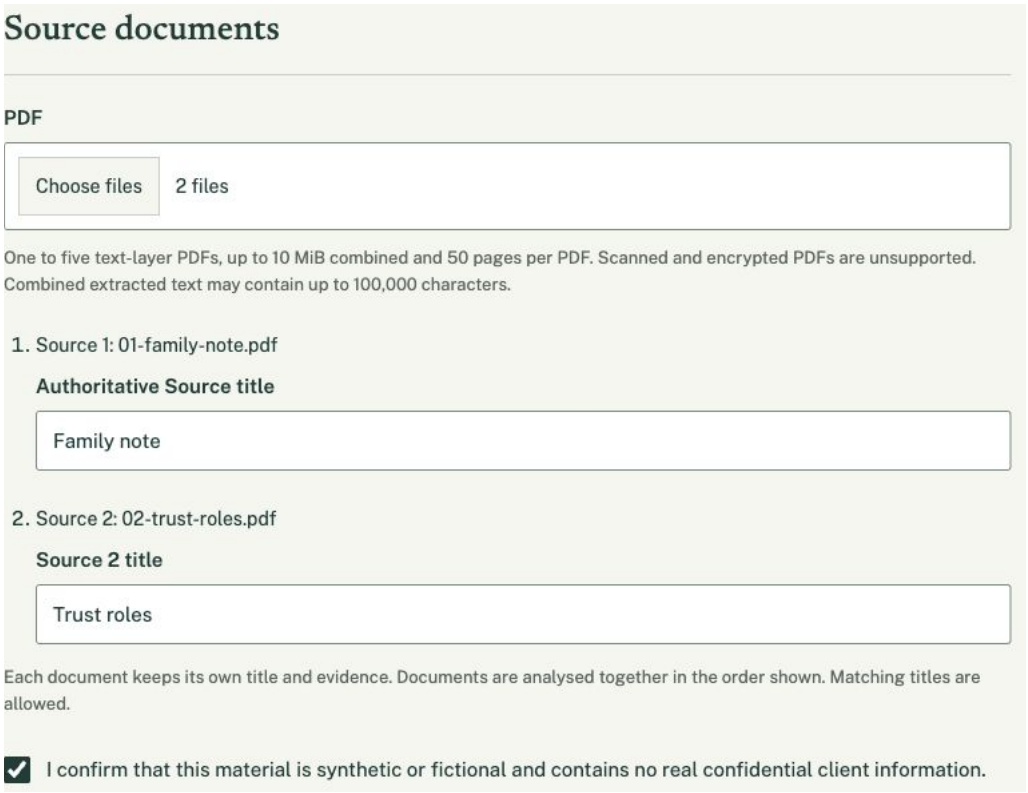
Source detail: Attendance Note - Meeting with Alice Chen

She was then asked about Bob Chen so that the relationship between the individuals being discussed was clear. Alice Chen confirmed that Alice Chen and David Chen are the parents of Bob Chen.

The quotation retains its Source identity. Selecting “View in source” reveals the passage in context; a highlighted sentence supports inspection but does not replace professional judgment.

Several PDFs.
One reviewable proposal.

An allowlisted signed-in user supplies a Matter reference, a title and one to five synthetic text-layer PDFs. Each document keeps its own Source identity and title. The documents are analysed together to produce one combined proposal.



Two distinct Sources

The recorded intake uses “Family note” and “Trust roles”. The Source titles and selected PDFs shown here were entered in the real application.

Historical authenticated UI capture. Synthetic inputs; Google and model boundaries substituted. This records the workflow, not extraction accuracy.

Deterministic PDF acquisition establishes the Authoritative Source text used for extraction, exact Evidence matching and later review. The raw PDFs are discarded after the intake request, including when processing fails; the reviewable record is the extracted text.

A proposal is saved only after successful analysis. It remains proposed state until the professional makes an explicit decision.

PDF acquisition: ADR 0005. Historical capture provenance in product-demo/.

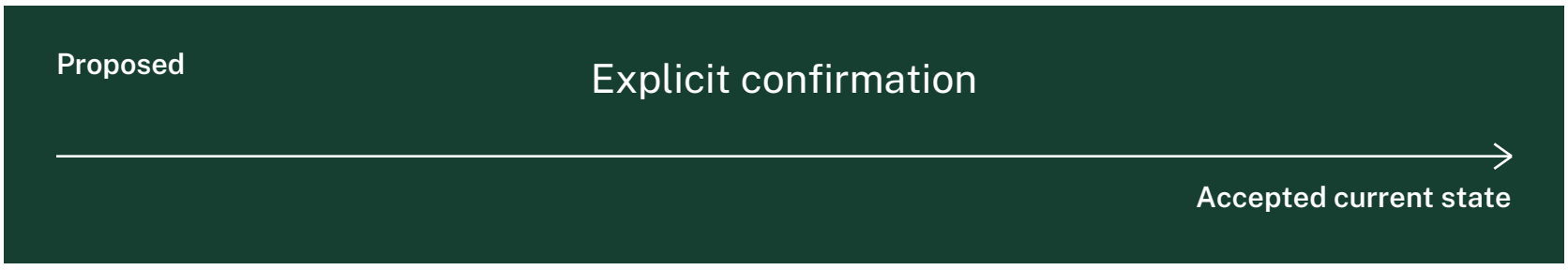
The decision belongs to the professional.

A mechanically valid graph is still a machine proposal. The saved proposal can be reopened for review; it does not silently become a Matter. The professional inspects the complete graph and supplied Sources, then confirms the whole proposal or discards the intake.

Actual whole-proposal decision



Historical authenticated capture, before confirmation. The local model was substituted; proposal confirmation and persistence used the real application paths.



Confirmation persists the entire proposed graph, all supplied Sources and Matter identity as professionally accepted current Matter state. In the recorded demonstration, the accepted Matter was reopened and its parent relationship inspected without reanalysis.

This is whole-proposal acceptance. Per-relationship editing and incremental updates to an existing Matter are outside this workflow.

Recorded transaction, confirmation and reopen: product-demo/PROVENANCE.md.

One relationship. More than one Source.

In authored Case 04, Andrew Chan's beneficiary relationship to the Hillcrest Family Trust appears in the first meeting and is confirmed at a follow-up. The relationship can carry Evidence from both Sources, with each quotation attributed to its own text.

Current UI detail: two supporting Sources



Source 2 · sources/note_02.txt · note_02_evidence_001

> The discussion returned first to the beneficiary position. Victoria Chan was asked to confirm the position concerning Andrew Chan. She confirmed that Andrew Chan is a beneficiary of the Hillcrest Family Trust.

Curated authored ideal in the current review workspace. This is synthetic demonstration data, not a model outcome or accepted Matter.

The second note supplements the first rather than revising it. Corroboration here does not demonstrate general conflict resolution, temporal reasoning or automatic updates to an accepted Matter.

Exact follow-up passage and Source identity: authored Case 04.

AI interprets.
Code enforces the rules.

Language requires interpretation. Graph integrity should not depend on another guess. PCG assigns those jobs to different parts of one fixed Extract / Resolve / Construct pipeline.

Extract relationships

Model

Read the ordered Sources together. Propose relationship candidates with named endpoints and Source-attributed quotations.

Resolve names

Configuration choice

Application intake currently uses no resolution. Research can add a constrained model stage to correct supported endpoint names.

Construct the graph

Deterministic code

Validate types, references and exact quote occurrence. Assign stable identities, pool Evidence and produce canonical graph structure.

Mechanical validity is only one boundary

Code can establish that a quotation occurs in a Source. That does not prove it supports the relationship. Professional acceptance remains explicit; deterministic benchmark evaluation runs separately.

Architecture: ADRs 0001 and 0009; current application configuration.

A case-led method, from hypothesis to product

The methodology starts with a documentary challenge. Development examples make the expected behaviour explicit; separate held-out examples test it. The Workbench makes the outcomes inspectable, and the surrounding application supports the workflow.

1

Author a case hypothesis

Define the new difficulty and what to hold stable, so a failure can be diagnosed.

2

Build development test cases

Write synthetic Source samples and explicit ground-truth relationships and Evidence.

3

Build a separate held-out test set

Create new synthetic situations and check their truth separately from authoring.

4

Measure and plot in the Workbench

Compare frozen runs, inspect relationship and Evidence scores, and trace errors to Sources.

5

Build the surrounding application

Support document intake, graph and Evidence inspection, and explicit professional acceptance.

Hackathon scope: baseline extraction plus constrained alias resolution. The following roadmap sets out the broader programme; it is not a list of completed experiments.

Author's methodology; case hypotheses, held-out records and Workbench implementation.

Twenty scenarios. One staged research programme.

We planned a progression of hypotheses about extraction: introduce a specific difficulty, diagnose the failure, then test an improvement. Within the hackathon window, we established the **baseline and a constrained alias resolver**. The wider programme remains ahead.

Planned challenges, in increasing complexity

01–05 have authored development cases. 06–20 are planned scenarios, not validated capabilities.

01 Explicit family / Trust facts

02 Dense facts and overlapping roles

03 Aliases for the same person

04 Cross-document reconciliation

05 Changed relationships

06 Contradictory Sources

07 Ambiguous role attribution

08 Indirect relationship descriptions

09 Negation and hypothetical claims

10 Larger, denser family graphs

11 Multiple Trusts

12 Corporate / non-person trustees

13 Different document genres

14 Locally resolvable pronouns

15 Sibling / cousin distinction

16 Overlapping Evidence spans

17 Longer Sources and distractors

18 Unnamed participants / omission

19 Historical versus current state

20 Combined stress test

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20 roadmap scenarios · 5 authored development cases · 10 separate held-out cases. These are different scopes. The held-out study evaluates the baseline-plus-resolver slice; it does not complete the twenty-scenario roadmap.

Case progression: cases/README.md. Scenario labels condensed for this document.

A small benchmark, with an explicit boundary

The first five roadmap scenarios supplied development cases. A separate held-out set tested whether those choices carried into new documentary situations. All cases were authored for this project and contain synthetic information.

10
held-out cases

14
Sources

39
expected relationships

What the held-out cases ask

Held-out 02 — **Dense wording**
Separate sibling status from beneficiary roles.

Held-out 05 — **A completed replacement**
Retain earlier facts; remove the superseded trustee.

Held-out 08 — **Matching document titles**
Preserve distinct Source identities.

Ground truth was checked separately from authoring by AI validators. These were internal checks, not external human or institutional validation. The retained comparison freezes its inputs and records its outcomes; it does not turn this small synthetic set into evidence of field performance.

Benchmark counts: retained publication; independent AI checks: validation records.

Count the relationships. Then check the Evidence.

The primary score compares exact relationship edges: the named starting entity, relationship type and named target entity. Direction matters for directed links; symmetric links are canonicalised. A plausible sentence alone is not a match.

$$F1 = 2 \times TP / (2 \times TP + FP + FN)$$

TP: matched edges · FP: extra edges · FN: missing edges

Two valid averages answer different questions

94.87%

Micro-F1, baseline A. Pool edge counts across all cases. Cases with more relationships contribute more edges.

93.33%

Macro-F1, baseline A. Average the ten case F1 scores. Each case contributes equally, whatever its size.

Provenance is assessed separately. Approved-quote checks concern matched relationships; Source/quote-pair checks concern evidence pairs. Their populations differ, and neither score should be read as another relationship-accuracy percentage.

Definitions: semantic_edges_v1; separate approved-provenance diagnostics.

Change one stage. Keep the extraction fixed.

A fresh extraction could change for reasons unrelated to resolution. The comparison therefore reuses the same saved model outputs and adds a zero-call replay control.

A Fresh baseline extraction

10 cases. One fresh model call per case.
Retain all extraction outputs for reuse.

A′ Exact replay

Saved extraction
No resolution
0 provider calls

B With resolver

Same saved extraction
Model resolves alternative names for the same person
10 additional provider calls

Same deterministic construction and evaluation

A′ reproduces A's outcomes. This helps isolate the resolver's effect from new extraction variability; it is **not an independent replication**. A′ and B use the same implementation and frozen inputs.

Retained A / A′ / B records; post-hoc exploratory comparison.

Two corrections explain the entire gain

On 10 project-authored synthetic held-out cases, relationship micro-F1 rose from **94.87% to 100%**. The change came from two endpoint-name corrections for one person in one case.



Actual published Workbench chart. Run 1 = A; Run 2 = A′; Run 3 = B. The full 0-1 F1 scale is preserved; points are not a time trend.

Pooled across 39 expected relationships	A / A′	B
Matched relationships	37	39
Extra / missing relationships	2 / 2	0 / 0

Evidence checks remained imperfect: B passed 36/39 approved-quote checks and 37/41 Source/quote-pair checks, using separate denominators. This small, post-hoc result does not establish general identity resolution or production accuracy.

One person.
Two corrected endpoints.

Heldout_03: factual interview note

> “Beatrice Ng attended the factual interview and asked to be called Bea. In this note, Bea means Beatrice Ng; no second person is being introduced.”

Exact Source excerpt from the synthetic held-out case. Editorial figure.

Endpoint correction			Unchanged target
Settlor of			
Bea	⟶	Beatrice Ng	Larch Family Trust

“Bea is the settlor of the Larch Family Trust.”

Endpoint correction			Unchanged target
Parent of			
Bea	⟶	Beatrice Ng	Isaac Ng

“Bea is the parent of Isaac Ng.”

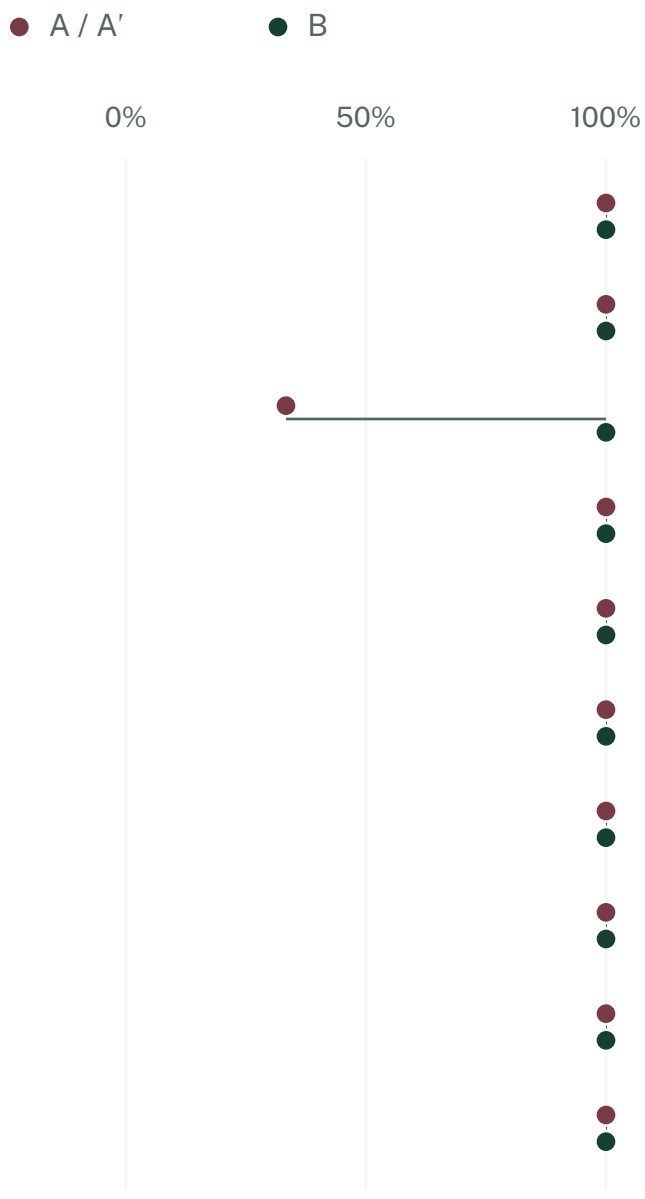
Exactly two endpoint names change from Bea to Beatrice Ng: the settlor of Larch Family Trust and the parent of Isaac Ng. Their quotations remain unchanged. The already-correct Isaac Ng beneficiary relationship also remains.

Retained heldout_03 Source, endpoint edits and relationship observations.

Nine cases already had perfect relationship F1

Every case except the explicit Bea short form scored 100% relationship F1 before resolution and after it. This locates the gain precisely; it does not demonstrate a broad improvement across ten distinct failures.

Relationship F1 per case



Case 03: 33.33% to 100%. Points use the full 0–100% scale. The paired markers are separated vertically so equal scores remain visible. A′ exactly reproduces A. One execution per case; no repeated-trial uncertainty estimate.

Case names and plotted F1 values: retained publication. Editorial chart.

Perfect edges do not mean perfect Evidence

B passed **36/39 approved-quote checks** and **37/41 Source/quote-pair checks**. These are separate populations. Exact approved boundaries can fail even when a shorter quotation supports the relationship.

Case 02: Grace Wong is a sibling of Hugo Wong

Approved span

“Felix Wong is the settlor of the Amber Family Trust, and Grace Wong and Hugo Wong are siblings and are both beneficiaries of the Amber Family Trust.”

Returned span · same Source, later start boundary

“Grace Wong and Hugo Wong are siblings and are both beneficiaries of the Amber Family Trust.”

Exact retained excerpts. This mismatch does not imply an unsupported sibling claim.

The incremental resolver usage

10 calls · 9,821 tokens

9,432 input + 389 output. The 3,456 cached input tokens are included, not additional. USD cost is unpriced in the retained record.

All calls use deepseek-flash. This post-hoc comparison uses one provider/model and one execution per case; it supplies no repeated-trial estimate of stability or end-to-end latency claim.

A useful finding.
A defined boundary.

The experiment demonstrates a traceable correction under controlled inputs. It also shows why an attractive graph and a high relationship score must remain open to inspection.

What the record supports

The resolver repaired the two observed endpoint names without changing the other nine cases' relationship outcomes. The retained examples also include multi-source corroboration, identical Source titles and explicit trustee replacement. These are observations within this dataset.

What it cannot establish

Ten short, project-authored synthetic cases are not a representative client-document collection. The study used one model, deepseek-flash, with one extraction and one resolver execution per case. There were no repeated trials or uncertainty estimates. The comparison was post-hoc; general identity resolution and real-document reliability remain untested.

An inspectable record, not a fresh model promise

Frozen Sources, settings, code identities, outputs, checksums and usage preserve the comparison. Replaying saved outputs reproduces the record, not a fresh model response. Workbench approval binds a frozen plan to a shared call budget; uncertain attempts remain counted and are not automatically retried.

The next version needs new evidence

The hackathon established baseline extraction and a constrained resolver. Continue the twenty-scenario programme by isolating new failure modes and testing improvements, alongside the product work below.

Validate the review workflow

Observe practitioners finding missing claims, checking Evidence and deciding whether to accept a proposal. Test the Trust layout and measure task outcomes before claiming time savings or usability.

Support change over time

Design incremental Matter updates, reviewed changes, corrections and historical state. A narrow trustee-replacement example does not yet provide a general temporal or contradiction-handling system.

Design the real-data boundary

Develop Matter authorization, tenancy and ownership, and explicit security, privacy, retention and data-handling requirements. Current sign-in and allowlisted actions do not authorize real client material.

Research should use newly unseen cases, richer documents and repeated runs. The published held-out set is now visible and must not be treated as unseen evidence in subsequent work informed by it.

Make every relationship reviewable

Private Client Graph combines a tangible practitioner workflow, explicit professional acceptance and a research record that can be inspected down to the original words. That is the foundation worth developing: AI-assisted structure whose claims remain open to challenge. The baseline and resolver are the first tested slice of a twenty-scenario research roadmap.

Open the prototype

private-client-graph-production.up.railway.app

Inspect the retained research

Research results and downloadable original publication

Read the implementation

github.com/alistairkung/private-client-graph

Sources and visual provenance

Product and architecture

README, glossary and ADRs 0001, 0005, 0006, 0009-0011.

Benchmark and retained results

Held-out cases, validator reports and the frozen A / A′ / B publication.

Product images

PR #160 current UI; PR #155 historical workflow captures. Synthetic fixtures; model substituted.

Research chart and editorial figures

Actual production chart captured 18 Oct 2026. Other diagrams explain recorded facts.

Synthetic-data prototype. Not validated for real client use.